

ECON 403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

Fall 2019

Department of Economics
Nazarbayev University

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Lecture time/Location: MW, 9:00 AM–10:15 AM, 8327

Office hours: By appointment, book weekly at <https://alejandromelo.youcanbook.me>, please sign at least one day in advance.

Course Description This course analyzes problems and issues created by informational asymmetries and how properly designed contracts could ameliorate those problems. In other words, this course takes a first look at the theory of incentives, which is at the core of modern microeconomics and game theory. Topics include adverse selection, screening, signaling, and moral hazard, as well as various applications to insurance, labor and credit markets, auctions, and organizational design. I will also introduce some formal instruments: how to work with uncertainty, equilibrium concepts in static and dynamic games and games with incomplete information. If time permits, we will look also at the basic theory of mechanism design.

Course Requirements ECON 201 Intermediate Microeconomics (C- and above) and ECON 301 Econometrics I (C- and above). The theory of incentives analyzes the design of optimal contracts and institutions in environments with asymmetric information and conflict of interest. The main mathematical tool used in the analysis is nonlinear optimization under constraints. We will briefly review the methods of constrained optimization if needed, but I will assume that you have a good background in calculus and probability. It is crucial that you are comfortable with analytical reasoning and mathematical notation in order to be able to follow the lectures and the readings.

Course Material There is no textbook required for this course. The course is based mainly on my lectures, slides and readings. All materials will be uploaded to Moodle.

The following is a list of undergraduate and graduate textbooks that you may find useful:

- Bolton and Dewatripont (2005), Contract Theory, MIT Press.
- Laffont and Martimort (2002), The Theory of Incentives, PUP.
- Salanie (1997), The Economics of Contracts: A Primer, MIT Press.
- Milgrom (2003), Putting Auction Theory to Work, Cambridge University Press.
- Krishna (2009), Auction Theory, Academic Press.

Grading Policy Grading will be weighted as follows:

For ECO403 (UG students)	For ECO503 (MA students)
Essay (12%)	Presentation (12%)
Problem Sets (20%)	Problem Sets (20%)
Midterm (30%)	Midterm (30%)
Final exam (35%)	Final exam (35%)
Presentation attendance (3%)	Presentation attendance (3%)

Problem Sets We will have 5 problem sets. Due: September 11, September 25, October 9, October 30, November 20.

Essay Undergraduate students have to write one essay where they describe some situation that happened to them, or that happened recently in the world or in some TV-series (or other piece of popular culture) that can be analyzed using one of the models we have studied. You have to write down a very simple model of this situation and then solve it. Finally, you need to comment about the results you got. Does the model predicts what happened in reality? Deadline: December 6

Midterm The midterm will be on October 16. It will cover roughly half of the material in the course.

Paper presentations Master students have to make paper presentations. Presentations will be on November 11 and 13. The papers will be distributed on September 16.

Presentation attendance All students should attend the paper presentations. I will record attendance on these days in Moodle.

Final Exam Will cover the second half of the material. The date will be chosen by the Registrar, during Final's Week.

A letter grade based on the sum of the components above will be assigned as follows:

Letter Grade	Score Achieved
A	[95, 100]
A–	[90, 95)
B+	[85, 90)
B	[80, 85)
B–	[75, 80)
C+	[70, 75)
C	[65, 70)
C–	[60, 65)
D+	[55, 60)
D	[50, 55)
F	[0, 50)

Course attendance Attendance is compulsory. Students who are recorded as absent in 14 or more class meetings will receive an automatic F grade. This includes excused absences. For example, 4 excused absences and 10 unexcused absences count as 14 absences and lead to letter grade F. You will be recorded as absent if you are late by more than 10 minutes or leave early, use your phone, chat with your friends, do work for other classes, or engage in any kind of disruptive behavior. It is your responsibility to keep track of your attendance record.

Make-up Policy For each regular exam, a make-up exam will be granted only once and only in case of properly documented emergencies, such as a medical report from the school medical center. Students who want to take a make-up exam need to inform me by email as early as possible but no later than the day after the regular exam. Make-up for the midterm exam will be given within a week after the exam date. Make-up for the final exam will be offered during after final's week and in accordance with the Registrar's policy. There will be no make-up for the make-up.

Rules and Regulations During lectures, please keep your laptops, tablets and cell phones switched off and refrain from text messaging. Do not arrive late or leave early without prior permission. Do not engage in side conversations or other disruptive behaviours.

Academic Misconduct Academic misconduct is an act in which a student:

1. seeks to claim credit for the work or efforts of another without authorization or citation (“plagiarism”);
2. uses unauthorized materials or fabricated data in any academic exercise;
3. forges or falsifies academic documents or records;
4. intentionally impedes or damages the academic work of others;
5. engages in conduct aimed at making false representation of any student’s academic performance;
6. engages in any form of cheating on an exam or assignment;
7. assists other students in any of these acts

These are serious offenses subject to disciplinary action depending on their severity as detailed in the Student Code of Conduct.

Tentative Course Outline

- Uncertainty. Expected utility, risky assets, risk aversion.
- Math review: constrained optimization.
- Asymmetric information: adverse selection, screening, competitive screening.
- Asymmetric information: signaling.
- Asymmetric information: moral hazard.
- Basic principal-agent model.
- Auctions, optimal auctions.
- Introduction to mechanism design.

Disclaimer Subsequent changes can be made to any part of this syllabus when necessary. Any changes will be announced in class and on Moodle.